

Опорные функции, сумма Минковского, полярность

Homework

1 сентября 2026 года

1. Prove that $\|\cdot\|_K$ is a norm exactly when K is a symmetric convex body, that the dual norm is $\|\cdot\|_{K^\circ}$, and that $K \subseteq L \iff L^\circ \subseteq K^\circ$.
2. (Cauchy 1841) Prove that the perimeter of a planar convex body equals $\int_0^\pi w(\theta) d\theta$, where $w(\theta)$ is the width in direction θ ; deduce Barbier's theorem (1860): a curve of constant width d has perimeter πd .
3. Prove that the mean width is additive: $w(K + L) = w(K) + w(L)$; compute it for the ball, the cube and the regular simplex.
4. Prove that the difference body $DK = K - K$ has $h_{DK}(u)$ equal to the width of K in direction u , that DK is symmetric, and that $DK = 2K$ if and only if K is symmetric.
5. (classical) Prove that a convex body in $\mathbb{R}^3$ all of whose orthogonal projections onto planes are discs is a ball. What happens if «disc» is replaced by «centrally symmetric set»?
6. (Mexmat, 2008) Prove that there exists a bounded convex figure in the plane without a centre of symmetry such that every line bisecting its perimeter also bisects its area.
7. (MIPT 2019) Suppose $2m$ points lie in the unit ball of a Euclidean space. Prove that they can be split into two groups of m points whose centres of mass are at distance at most $2/m$.
8. (ФКН, весна 2023) For a convex polytope $P \subseteq \mathbb{R}^d$ with the origin in its interior the polar polytope is $P^\Delta = \{x : \langle p, x \rangle \leq 1 \ \forall p \in P\}$.

Prove that for every d -dimensional convex polytope Q one can choose an interior point c such that the sum of the vertex vectors of $(Q - c)^\Delta$ equals zero.

9. (SEEMOUS 2008) Let $P_0, P_1, \dots$ be convex polygons where the vertices of P_{k+1} are the midpoints of the sides of P_k . Prove that there is a unique point lying inside all of them.
10. (ФКН, осень 2023) Let $B \subseteq \mathbb{R}^n$ be a bounded convex body with nonempty interior and let $M: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear map with $M(B) \subseteq B$. Prove that all eigenvalues of M , including complex ones, have modulus at most 1.
11. Prove that $\text{vol}(K) \text{vol}(K^\circ)$ is invariant under invertible linear maps; compute it for the ball, the cube and the cross-polytope and say which is largest.
12. Prove that volume and surface area are continuous with respect to the Hausdorff metric on convex bodies, and give an example showing that the number of vertices is not.